

Teaching fractions toolkit

Overview

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This document provides an overview of the Teaching Fractions Toolkit. The toolkit is based on the recommendations in [Developing Effective Fractions Instruction for Kindergarten Through 8th Grade](#), a What Works Clearinghouse practice guide. The toolkit supports implementation of the evidence-based recommendations in the practice guide. It provides resources for administrators and leaders (including principals, assistant superintendents, curriculum directors, mathematics coaches, and teacher leaders) as they address fraction instruction in kindergarten through eighth grade. The professional development modules, as described on the next page and intended to be led by a local facilitator, are for sixth-grade teachers to support their implementation of the practice guide recommendations.

To support use of the practice guide, the Teaching Fractions Toolkit includes the following:

- Three videos—one to introduce the toolkit and two to introduce what the practice guide recommendations look like in practice
- Two leader handouts—one summarizing the practice guide recommendations and one outlining the progression of fraction content represented in the practice guide
- Six professional development modules, with supporting resources, for grade 6 teachers
- Facilitation guides for school leaders to lead professional development for grade 6 teachers
- A tool for administrators and leaders to assess district/school conditions to support fractions instruction
- Links to existing resources that support use of the practice guide recommendations

The **leader** materials are designed to support administrators and mathematics leaders as they do the following:

Understand the evidence base for the recommendations

Mitigate roadblocks to recommendation implementation

Support implementation of the professional development modules with teachers

Assess and monitor relevant school and district conditions

Teacher professional development modules

The recommendations in the practice guide focus on the upper elementary and middle grades. The Teaching Fractions Toolkit professional development modules are for sixth-grade mathematics teachers because sixth grade is a high-leverage grade for student learning within this range. The professional development modules and facilitation materials focus on Recommendations 2, 3, and 4 and align with Recommendation 5, which states that professional development programs should place a high priority on improving teachers' understanding of fractions and how to teach them.

Professional development modules

Module 1

Fractions on a Number Line focuses on Recommendation 2, with a particular emphasis on number lines.

Module 2

Understanding Fraction Addition and Subtraction focuses on Recommendation 3 and uses visual models and formative assessments to identify misunderstandings.

Module 3

Understanding Fraction Multiplication and Division continues work on Recommendation 3. It includes computation in real-world contexts and judging the reasonableness of answers.

Module 4

Implementation of Fraction Computation focuses on approaches related to Recommendations 2 and 3.

Module 5

Understanding Ratio, Rate, and Proportion focuses on Recommendation 4 and includes visual models and apps.

Module 6

Implementation of Ratio, Rate, and Proportion focuses on approaches related to Recommendation 4.

Supporting resources

Teacher reflection tool to assess initial and developing classroom practices



Formative assessments to identify student misconceptions related to fractions



Mathematics tasks



Resources related to the practice guide recommendations



Classroom implementation tools and resources



Lesson planning frameworks



Videos



Approaches to remediate roadblocks to implementing the practice guide recommendations



Online apps with visual models



Terminology: The practice guide and the Teaching Fractions Toolkit materials use the term “fractions” rather than the term “rational numbers.” The term “fractions” refers to the full range of ways of expressing rational numbers, including decimals, percentages, and negative fractions.